

# LAB 7: BANKER'S ALGORITHM

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### C PROGRAM

```
#include <stdio.h>
int main()
{
    int n, m, i, j, k;
    int allocation[10][10], max[10][10];
    int need[10][10];
    int available[10];
    int finish[10] = {0};
    int safeSeq[10];
    int count = 0;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);
    printf("Enter Number of Resources: ");
    scanf("%d", &m);

    printf("
Enter Allocation Matrix:
");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &allocation[i][j]);

    printf("
Enter Maximum Matrix:
");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &max[i][j]);

    printf("
Enter Available Resources:
");
    for(i = 0; i < m; i++)
        scanf("%d", &available[i]);

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            need[i][j] = max[i][j] - allocation[i][j];
        }
    }

    while(count < n)
    {
        int found = 0;
        for(i = 0; i < n; i++)
        {
            if(finish[i] == 0)
            {
```

```

        for(j = 0; j < m; j++)
        {
            if(need[i][j] > available[j])
                break;
        }
        if(j == m)
        {
            for(k = 0; k < m; k++)
                available[k] += allocation[i][k];
            safeSeq[count++] = i;
            finish[i] = 1;
            found = 1;
        }
    }
}
if(found == 0)
{
    printf("
System is NOT in Safe State
");
    return 0;
}

printf("
System is in Safe State
");
printf("Safe Sequence: ");
for(i = 0; i < n; i++)
    printf("%d ", safeSeq[i]);
printf("
");
return 0;
}

```

## SHELL SCRIPTING

```

#!/bin/bash
echo "Enter Number of Processes:"
read n
echo "Enter Safe Sequence (space separated):"
read -a seq
echo "Safe Sequence is:"
for ((i=0; i<n; i++))
do
    echo -n "P${seq[$i]} "
done
echo
echo "System is in Safe State"

```

## OUTPUT

```

(kali@Harish)-[~]
└─$ nano banker.sh
(kali@Harish)-[~]
└─$ chmod +x banker.sh
(kali@Harish)-[~]
└─$ bash banker.sh
Enter Number of Processes:

```

```
2
Enter Safe Sequence (space separated):
3
Safe Sequence is:
P3
System is in Safe State
```